Distance Measuring Equipment (DME) (Mod 6/2049) – General Data

1. General

The Distance Measuring Equipment (DME) gives the distance from the aircraft to a DME ground station. The DME consists of two transceivers (DME 1 and DME 2) and one antenna per DME transceiver. DME 1 antenna and DME 2 antenna. Provision has been made for the DMEs, DME 1 and DME 2 to be installed on the aircraft, this is part of the IFR Operations Bundle customer order option 6209.

For aircraft incorporating customer Standard Order Option S.O.O. 6388 there is one DME transceiver (DME 1) and one antenna installed on the aircraft.

For aircraft with S.O.O. 6388 incorporated, the second DME transceiver (DME 2) and associated antenna are installed on the aircraft by incorporating customer Standard Order Option S.O.O. 6386.

2. Description

KN 63 Distance Measuring Equipment (DME) transceivers may be installed in the rear avionics rack. The transceiver has 200 different channels. The transmitter processes signals between 1,025 MHz and 1,150 MHz and the receiver processes signals between 962 MHz and 1,213 MHz. Range to station are output to the MAU as digital serial data. The maximum displayed range of the DME transceiver is 300 nautical miles. The DME detail window can be displayed in the radio tuning window by pressing the DME button on the PFD Controller. An alternative means to accessing the DME window is by using the DME Detail soft key in the NAV detail window. The DME detail window contains soft keys ‘DME PAIR' to select the association of the DME to NAV 1 or NAV 2, and ‘DME HOLD' to select DME hold ON or OFF. When the DME HOLD is selected to ON, an H adjacent to the DME distance is displayed on the PFD HSI display.

DME 1 antenna is installed on the bottom of the fuselage, left of the centre line, between station 291.00 and 301.82. DME 2 is located on the lower empennage FS431 between the two Radar Altimeter antennas. The DME antenna is a blade shape and has a low contour with an antenna connector on the base.

When the system transmits, a suppression pulse is sent to the ATC transponder system and the TCAS to prevent operation of these systems at the same time. The DME receives a signal with an identification code from the ground station. The signal is sent to the Audio Control Panel, which generates a tone to identify the ground station.

28 Volt DC power is protected by a 3 Amp DME 1 (Left Bus) and DME 2 (Right Bus) circuit breaker is located on the avionics circuit breaker panel (refer to 24-00-00, Electrical Power – General Data).

The KN 63 Distance Measuring Equipment System interfaces to the following aircraft systems:

A. Modular Avionics Unit (MAU) (refer to 31-43-00, Modular Avionics Unit (MAU) (Mod 6/2049) – General Data).

B. Dual KMA 29 Audio Control Panel (refer to 23-53-00, Audio Control Panel – General Data).

C. A suppression pulse to/from the Single or Dual (if installed) Transponder System (refer to 34-53-00, KXP 2290 Transponder System – General Data).

D. A suppression pulse to/from the TCAS (refer to 34-43-00, Traffic Alert and Collision Avoidance System I (TCAS I) – General Data and 34-43-01, Traffic Alert and Collision Avoidance System II (TCAS II) (Mod S.O.O. 6219) – General data), if installed.

3. Adjustment/Tests

A. Function Test of Distance Measuring Equipment

1) General

These procedures are for the function test of DME 1 and DME 2 transceivers (KN 63).

2) Function Test

a) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

b) Wait several minutes for the aircraft electrical and avionics systems to initialize.

c) Ensure that DME 1 (H3) (Left Bus) and/or DME 2 (H4) (Right Bus), circuit breaker on the avionics circuit breaker panel are closed.

d) In the VHF NAV detail window select the DME detail page. The DME detail page can also be accessed by pressing the DME button on the PFD/Radio controller.

e) Channel the NAV 1 and NAV 2 frequency to a local ground station. Station ident will be heard on the aircraft audio panel.

f) Ensure a valid DME 1 and/or DME 2 indication is on the LH PFD and RH PFD.

g) With a valid DME 1 and/or DME 2 indication; select DME hold to ON and verify an amber H is displayed on both the LH PFD and RH PFD.

h) Channel the associated NAV 1 and NAV 2 to a invalid DME frequency. Verify DME 1 and/or DME 2 distance continues to be displayed.

i) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

B. Electrical Bonding Test – DME 1 Antenna

1) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00-00, Electrical Power – General Data).

2) Gain access to Access and Inspection Provision 33 panel. (Refer to 12-00-00, Servicing – General Data).

3) Remove screws and remove access panel.

4) Carefully cut the sealant from around the base of the antenna and mounting screws.

5) Verify that the electrical conductivity is consistent over the electrical bond contact area by placing a probe on the DME antenna mounting screw head. Place the second probe on an anchor nut from the Access and Inspection Provision 33 panel. The resistance between the two points must be < 3.0 milliohms unless stated otherwise.

6) Check the electrical conductivity between several random mounting screws within the electrical bond contact area to ensure consistency.

7) If the electrical conductivity is not consistent over the electrical bond contact area check the antenna electrical bonding in accordance with 20-75-00, Electrical Bonding and Grounding of Aircraft Structures – General data.

8) Place Access and Inspection Provision 33 panel onto access cavity and tighten the retaining screws.

9) Seal around the antenna base and dome seal the securing screws. (Refer to 20-80-00, Installation of Aircraft Antennas – General data).

C. Electrical Bonding Test – DME 2 Antenna (if installed)

1) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

2) Gain access to empennage through rear baggage compartment.

3) Carefully cut the sealant from around the base of the antenna and mounting screws.

4) Verify that the electrical conductivity is consistent over the electrical bond contact area by placing a probe on the DME antenna mounting screw head. Place the second probe on the nearest ground stud. The resistance between the two points must be < 3.0 milliohms unless stated otherwise.

5) Check the electrical conductivity between several random mounting screws within the electrical bond contact area to ensure consistency.

6) If the electrical conductivity is not consistent over the electrical bond contact area check the antenna electrical bonding in accordance with 20-75-00, Electrical Bonding and Grounding of Aircraft Structures – General data.

7) Seal around the antenna base and dome seal the securing screws. (Refer to 20-80-00, Installation of Aircraft Antennas – General data).

4. Removal/Installation

A. Removal/Installation of DME Transceiver (KN 63)

1) General

These procedures are for the removal and installation of the DME 1 transceiver and DME 2 transceiver.

2) Removal

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00-00, Electrical Power – General Data).

b) Pull circuit breaker DME 1 (H3) (Left Bus), DME 2 (H4) (Right Bus) respectively on the avionics circuit breaker panel and fit safety clips.

c) Gain access to the aft avionic shelf through the rear baggage compartment.

d) Disconnect the electrical and coaxial cables at the front of the DME transceiver unit.

e) Loosen the hold down clamp at the front of unit.

f) Remove unit from mounting tray assembly.

Note Stow connectors of the removed components.

g) Visually inspect mounting tray for condition, if damaged replace as per Para 4.B.

3) Installation

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00-00, Electrical Power – General Data).

b) Ensure the DME 1 (H3) (Left Bus), DME 2 (H4) (Right Bus) circuit breaker on the avionics circuit breaker panel are pulled and fitted with safety clips.

c) Gain access to the aft avionic shelf through the rear baggage compartment.

d) Place unit on the mounting tray assembly.

e) Slide unit onto mounting tray, ensure unit is correctly oriented on the mounting tray; electrical connections facing forward until the unit is snug and secure under the mounting tray lip.

f) Tighten the hold down clamps on the DME mounting tray. Ensure unit is securely held on mounting tray.

g) Reconnect the electrical and coaxial cables. Ensure connections are sound and secure.

h) Remove safety clips and replace circuit breaker DME 1 (H3) (Left Bus), DME 2 (H4) (Right Bus) respectively on the avionics circuit breaker panel.

i) Do a function test on the DME as per Para 3.A.2).

B. Removal/Installation of DME Transceiver Mounting Tray

1) General

These procedures are for the removal and installation of the DME 1 transceiver mounting tray and the DME 2 transceiver mounting tray.

2) Removal

a) Remove DME transceiver as Para 4.A.2).

b) Remove the four lock nuts and washers from the bottom of the shelf that secure mounting tray to shelf.

c) Remove mounting tray from aircraft.

3) Installation

a) Locate mounting tray on avionic shelf, align mounting holes, ensure the hold down clamp is forward.

b) Install the four lock washers and nuts from the bottom of shelf and ensure tray is securely held on shelf.

C. Removal/Installation DME 1 Antenna and DME 2 Antenna (if installed)

1) General

These procedures are for the removal and installation of DME 1 and DME 2 antenna.

2) Removal

a) Ensure the aircraft power is OFF. (Refer to 24-00-00-00, Electrical Power – General Data).

b) Pull the DME 1 (H3) (Left Bus) and DME 2 (H4) (Right Bus) circuit breakers on the avionics circuit breaker panel and fit safety clips.

c) Carefully cut the sealant from around the base of the antenna and mounting screws.

d) Support the antenna and remove the four screws from the base of the antenna.

e) Carefully lift the antenna clear of the fuselage until there is access to the coaxial connection.

f) Disconnect the coaxial connector.

g) Remove antenna from Aircraft.

Note Stow connectors of the removed components.

3) Installation

WARNING **INSTALLATION OF THIS COMPONENT CONTAINS PROTECTION FEATURES THAT ARE CRITICAL TO PREVENTING DEGRADATION FROM THE EFFECTS OF HIGH INTENSITY RADIATED FIELDS (HIRF) AND INDIRECT EFFECTS OF LIGHTNING (IEL). ENSURE INTEGRITY OF THE HIRF/IEL PROTECTION FEATURES ARE MAINTAINED IN ACCORDANCE WITH THE FOLLOWING:**

CHASSIS GROUNDING – ALL RACK MOUNT / REMOTE MOUNT UNITS SHALL BE ELECTRICALLY BONDED TO THE AIRFRAME. ENSURE THAT THE MATING SURFACES BETWEEN THE LRU MOUNTING TRAY (OR LRU MOUNTING FEET IF A TRAY IS NOT USED) PROVIDE A LOW IMPEDANCE ELECTRICAL PATH. ALL MATING SURFACES MUST BE FREE OF ALL PAINT AND OTHER NONCONDUCTIVE ELEMENTS. REFER TO BONDING REQUIREMENTS IN 20-75-00, ELECTRICAL BONDING AND GROUNDING OF AIRCRAFT STRUCTURES – GENERAL DATA.

SHIELD GROUNDS – VERIFY SHIELD GROUNDING AT CONNECTOR BACKSHELLS AS PER REQUIREMENTS IN PSM 1-64-2W.

BACKSHELLS OR STRAIN RELIEFS – VERIFY PROPER ELECTRICAL BONDING BETWEEN THE BACKSHELL/STRAIN RELIEF AND THE MOUNTING TRAY OR LRU CASE AS PER REQUIREMENTS IN PSM 1-64-2W.

a) Ensure the aircraft power is OFF. (Refer to 24-00-00-00, Electrical Power – General Data).

b) Ensure that the DME 1 (H3) (Left Bus) and DME 2 (H4) (Left Bus) circuit breakers on the avionics circuit breaker panel are pulled and fitted with safety clips.

c) Solvent clean the electrical contact bond areas immediately before the installation. (Refer to 20-75-00, Electrical Bonding and Grounding of Aircraft Structures – General data).

d) Bring the antenna assembly close to the fuselage, connect the coaxial connector.

e) Secure the antenna assembly on fuselage with four mounting screws.

f) Solvent clean the exposed area of the airframe around the antenna. (Refer to 20-75-00, Electrical Bonding and Grounding of Aircraft Structures – General data).

g) Touch up the exposed area of airframe around the antenna if required with F19 primers by brush within 24 hours of the antenna installation. Ensure that the primer re-finishing provides complete coverage of the affected area.

h) If the affected area requires paint touch up, brush coat the area with the top-coat paint system matching the existing.

i) Seal around the antenna base and dome seal the securing screws. (Refer to 20-80-00, Installation of Aircraft Antennas – General data). The aircraft will be ready to fly approximately 24 hours after sealing.

j) Check the antenna electrical bonding as per Para 3.B for DME 1 and Para 3.C for DME 2.

k) Remove the safety clips and replace DME 1 (H3) (Left Bus) and DME 2 (H4) (Right Bus) circuit breakers on the avionics circuit breaker panel.

l) Do a function test on the DME system as per Para 3.A.2).

5. Troubleshooting

A. DME 1 and DME 2

1) If the DME is deemed inoperative check the following:

a) Ensure DME 1 (H3) and/or DME 2 (H4) circuit breaker are closed.

b) Visually inspect DME connector for damaged wires or contacts.

c) Visually inspect DME antenna for corrosion and contaminants.

d) Ensure that the TCAS (refer to 34-43-00, Traffic Alert and Collision Avoidance System I (TCAS I) – General Data) and Transponder (34-43-01, Traffic Alert and Collision Avoidance System II (TCAS II) (Mod S.O.O. 6219) – General data) suppression lines are functioning correctly, if installed.

e) Check DME antenna and cable connection for corrosion and contaminants. DME range can be affected with having a dirty (oil covered) antenna.

f) Ensure that the antenna coaxial cable connector are properly mated and secure.

g) Inspect antenna external surfaces for damage.

h) Check that the antenna is properly mounted and screws are tight.

i) If troubleshooting at the wire level is required refer to the applicable wiring diagrams in Wiring Manual PSM 1-64-2W.

j) If repair or service is required, send to appropriate rated avionics repair facility.